DRUGS · LOCAL ANAESTHETICS AND VASOCONSTRICTORS

Topical anaesthesia of the ear: phenol, EMLA and tetracaine

The ear drum and the bony ear canal are among the most sensitive surfaces in the body, and procedures on them in the clinic depend on good topical anaesthesia. Phenol cauterises and anaesthetises a small area of the drum in seconds; EMLA and tetracaine numb the drum and canal over 30 to 60 minutes; and injection of local anaesthetic into the canal (a four-quadrant block) anaesthetises the whole canal and drum.

1 CHAPTER ONE At a glance

The agents on one page.

1.1 Summary

TOPICAL ANAESTHESIA OF THE EAR

	SUMMARY
Agents	Phenol (liquefied, about 80% to 90%) applied to a spot; EMLA cream (lidocaine 2.5% and prilocaine 2.5%) filling the canal; tetracaine (amethocaine) 4% gel or solution; lidocaine infiltration of the canal
ENT uses	Myringotomy and grommet in the clinic; intratympanic injections; biopsy of the drum; removal of granulation or small polyps; clinic myringoplasty
Phenol	Apply a tiny amount with a fine applicator to a clean, dry drum at the planned site; the drum blanches white within seconds and the area is anaesthetic for minutes
EMLA	Fill the canal to the drum, leave for 30 to 60 minutes, then suction out completely
Tetracaine	Fill the canal for 15 to 45 minutes, then remove

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Choosing

Phenol is quick and precise for a single puncture or a small incision. EMLA and tetracaine cover more of the drum for longer procedures but take time and must be removed fully. Injecting the canal at the bony-cartilaginous junction in four quadrants with lidocaine and adrenaline numbs both canal and drum and reduces bleeding.

2.2 Safety with a perforation

Topical anaesthetics reaching the middle ear can cause **transient vertigo** or facial weakness; avoid filling an ear with a perforation.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Phenol: excess causes a burn and a persistent perforation. EMLA: **methaemoglobinaemia** from prilocaine, mainly in infants and with large areas (not a practical concern in the adult ear). Tetracaine: allergy (ester anaesthetic). All: vertigo if they reach the inner ear.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- How to anaesthetise a drum for a clinic grommet.
- The four-quadrant canal block.

4.2 Common pitfalls

COMMON PITFALLS

- Using too much phenol.
- Leaving EMLA in the canal.

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